

Bhavin K. Moriya

Data Scientist | Machine Learning | Privacy-Preserving Analytics

EU Blue Card Holder (Germany)

bhavinmoriya58@gmail.com

Google Scholar GitHub LinkedIn Kaggle YouTube RoutingApp StreamlitApps



Experience

Data Scientist / Research Assistant	Hochschule Esslingen, Germany	Nov 2023 – Dec 2025
-------------------------------------	-------------------------------	---------------------

- **AnoMoB Project:** Developed first-of-its-kind encrypted comparison operations and non-linear function evaluations using **OpenFHE (C++)** and **TFHE-rs (Rust)**.
- **Sustainability Analysis:** Leveraged **SUMO** to simulate movement data and conducted comparative analyses of travel modalities focusing on **CO2 emissions** and optimal route duration.
- **Decision Support for Sustainable Mobility:** Developed a **decision-support tool** for the ANOMOB project that uses a weighted scoring system to help users evaluate the trade-offs between travel time and CO2 emissions. It details a technical architecture based on Dockerized routing engines (ORS/OTP) and highlights research in privacy-preserving mobility analytics and homomorphic encryption.
- **Performance Benchmarking:** Co-authored breakthrough research comparing runtime and memory consumption of **CKKS** and **TFHE** schemes, published in Cryptology ePrint Archive (2025).
- **Data Engineering:** Built production-quality software tools for statistical analysis of mobility data using **Python**, **polars**, and **geopandas**.

Assistant Professor	Univ. of Viçosa, Brazil	Nov 2015 – Nov 2023
---------------------	-------------------------	---------------------

- Taught undergraduate and master's level mathematics; supervised 3 dissertations.
- Collaborated with international researchers, resulting in 16+ peer-reviewed publications.

Postdoctoral Fellowships

- IMSc (2011–2013), HRI (2013–2014), CMI (2014), Univ. of Brasília (2014–2015)

Education

- | | |
|--|-------------|
| • Ph.D. in Mathematics , Harish Chandra Research Institute
Thesis: Some Zero Sum Problems in Combinatorial Number Theory | 2011 |
| • M.Sc. & B.Sc. in Mathematics , The M.S. University of Baroda | 2003 – 2005 |

Skills

Programming: Python (Proficient), C++, Rust, SQL, $\LaTeX$, Go, Julia

Data & Simulation: polars, geopandas, SUMO, Scikit-learn, Docker

Workflow: LLM-assisted development (GitHub Copilot), Git, Agile Research, Streamlit

Languages: English (Fluent), German (**B1/Professional Communication**), Portuguese (Fluent/Professional), Hindi (Native), Gujarati (Native)

Selected Projects (GitHub)

- **Credit Risk Model — PD • LGD • CCF**
- **Vascular Surrogate Models** – Accelerating Intracranial Aneurysm CFD.
- **AudioMNIST** – A spoken digit recognition system.

Publications & Awards

- (2025) **A Performance Comparison of Homomorphic Encryption Schemes CKKS and TFHE**, Cryptology ePrint Archive.
- 16 peer-reviewed publications in combinatorics and cryptography.
- Recipient of multiple international grants (ICM, CIMPA-ICTP, FAPEMIG).